Practice Questions for Duality Theory and Applications

Generated by Scribe.AI

December 31, 2024

1 Questions

1. In the context of linear programming duality, what is the significance of the Separating Hyperplane Theorem, and how does it relate to the feasibility and boundedness of primal and dual problems?

   a. The Separating Hyperplane Theorem is irrelevant to linear programming duality.

   b. The theorem states that if a primal problem is infeasible, its dual must be unbounded, and vice versa.

   c. The theorem provides a geometric interpretation of strong duality; if both primal and dual problems are feasible, there exists a hyperplane separating their feasible regions.

   d. The theorem guarantees the existence of an optimal solution for both the primal and dual problems, regardless of feasibility.

   e. The theorem is only applicable to primal problems with a bounded feasible region.

2. Consider a primal linear programming problem and its dual. The primal problem is found to be infeasible. Using the Separating Hyperplane Theorem, what can we conclude about the dual problem?

   a. The dual problem is also infeasible.

   b. The dual problem is unbounded.

   c. The dual problem has a unique optimal solution.

   d. The dual problem is feasible but unbounded.

   e. We cannot conclude anything about the dual problem.

3. Suppose we have a primal linear programming problem and its dual. Both problems are feasible. What does strong duality tell us about the relationship between their optimal objective function values?

   a. The optimal objective function value of the primal problem is less than or equal to that of the dual problem.

   b. The optimal objective function value of the primal problem is greater than or equal to that of the dual problem.

   c. The optimal objective function values of both problems are equal.

   d. There is no relationship between the optimal objective function values.

   e. The optimal objective function value of the primal problem is strictly less than that of the dual problem.

4. Explain the concept of complementary slackness in linear programming and its significance in verifying the optimality of solutions. Illustrate your explanation with a simple example.

   a. Complementary slackness is a concept unrelated to linear programming optimality.

   b. Complementary slackness states that if a primal constraint is binding, the corresponding dual variable must be zero, and vice versa.

   c. Complementary slackness states that if a primal variable is positive, the corresponding dual constraint must be binding, and vice versa.

   d. Complementary slackness is only applicable to primal problems with equality constraints.

   e. Complementary slackness is a necessary but not sufficient condition for optimality.

5. Consider a primal linear programming problem and its dual. Explain the relationship between the feasibility and boundedness of the primal and dual problems, covering all four possible scenarios. For each scenario, describe the implications for the optimal objective function values of both problems.

   a. If the primal is feasible and bounded, the dual is infeasible. If the primal is infeasible, the dual is unbounded. If the primal is unbounded, the dual is infeasible. If the primal is infeasible and unbounded, the dual is feasible and bounded.

   b. If the primal is feasible and bounded, the dual is feasible and bounded, and their optimal objective function values are equal. If the primal is infeasible, the dual is either infeasible or unbounded. If the primal is unbounded, the dual is infeasible. If the primal is infeasible and unbounded, the dual is infeasible.

   c. If the primal is feasible and bounded, the dual is feasible and bounded, and their optimal objective function values are equal. If the primal is infeasible, the dual is unbounded. If the primal is unbounded, the dual is infeasible. If the primal is infeasible and unbounded, the dual is feasible and bounded.

   d. If the primal is feasible and bounded, the dual is feasible and bounded, and their optimal objective function values are equal. If the primal is infeasible, the dual is unbounded. If the primal is unbounded, the dual is infeasible. If the primal is infeasible, the dual is either infeasible or unbounded.

   e. If the primal is feasible and bounded, the dual is infeasible. If the primal is infeasible, the dual is feasible and bounded. If the primal is unbounded, the dual is infeasible. If the primal is infeasible and unbounded, the dual is unbounded.

6. Given the primal problem: Maximize $Z = 2x_1 + 3x_2$ subject to $x_1 + x_2 \leq 5$, $x_1 \leq 3$, $x_2 \leq 4$, $x_1, x_2 \geq 0$. What is the objective function of its dual problem?

   a. Minimize $W = 5y_1 + 3y_2 + 4y_3$

   b. Maximize $W = 5y_1 + 3y_2 + 4y_3$

   c. Minimize $W = y_1 + y_2 + y_3$

   d. Maximize $W = 2y_1 + 3y_2$

   e. Minimize $W = 3y_1 + 2y_2$

7. For the primal problem: Maximize $Z = 3x_1 + 2x_2$ subject to $-x_1 + 3x_2 \leq 12$, $x_1 + x_2 \leq 8$, $2x_1 - x_2 \leq 10$, $x_1, x_2 \geq 0$, what are the constraints of its dual problem?

a. $-y_1 + y_2 + 2y_3 \geq 3$, $3y_1 + y_2 - y_3 \geq 2$, $y_1, y_2, y_3 \geq 0$

b. $-y_1 + y_2 + 2y_3 \leq 3$, $3y_1 + y_2 - y_3 \leq 2$, $y_1, y_2, y_3 \geq 0$

c. $-y_1 + y_2 + 2y_3 \geq 3$, $3y_1 + y_2 - y_3 \geq 2$

d. $-y_1 + y_2 + 2y_3 \leq 12$, $y_1 + y_2 - y_3 \leq 8$, $2y_1 - y_2 \leq 10$, $y_1, y_2, y_3 \geq 0$

e. $-y_1 + y_2 + 2y_3 \geq 12$, $y_1 + y_2 - y_3 \geq 8$, $2y_1 - y_2 \geq 10$, $y_1, y_2, y_3 \geq 0$

8. In the context of weak duality, if Z represents the objective function value of the primal problem and W represents the objective function value of the dual problem, which inequality always holds for a maximization primal problem?

   a. $Z \geq W$

   b. $Z \leq W$

   c. $Z = W$

   d. $Z > W$

   e. $Z < W$

9. Suppose the primal problem has an unbounded objective function. What can we conclude about the dual problem based on weak duality?

   a. The dual problem is unbounded.

   b. The dual problem is infeasible.

   c. The dual problem has an optimal solution.

   d. The dual problem is feasible but unbounded.

   e. There is no information about the dual problem.

10. Consider the primal problem: Maximize $Z = 3x_1 + 2x_2$ subject to $-x_1 + 3x_2 \leq 12$, $x_1 + x_2 \leq 8$, $2x_1 - x_2 \leq 10$, $x_1, x_2 \geq 0$. Formulate its dual problem. What is the constraint in the dual problem corresponding to the primal variable x_1?

    a. $-y_1 + y_2 + 2y_3 \leq 3$

    b. $-y_1 + y_2 + 2y_3 \geq 3$

    c. $y_1 + y_2 - y_3 \leq 3$

    d. $y_1 + y_2 + 2y_3 \geq 2$

    e. $y_1 + y_2 + 2y_3 \leq 2$

2 Answers

1. In the context of linear programming duality, what is the significance of the Separating Hyperplane Theorem, and how does it relate to the feasibility and boundedness of primal and dual problems?

   a. The Separating Hyperplane Theorem is a cornerstone of duality theory, providing a geometric understanding of the relationship between primal and dual problems.

   b. The Separating Hyperplane Theorem is crucial in duality theory. It states that if the primal problem is infeasible, then the dual problem is unbounded (or infeasible), and vice versa. This provides a powerful link between the feasibility of the primal and dual problems.

   c. While the theorem does relate to strong duality, it doesn't state that a hyperplane *separates* the feasible regions if both are feasible. Instead, it implies that if one is infeasible, the other is unbounded, and if both are feasible, their optimal objective values are equal.

   d. The theorem doesn't guarantee the existence of optimal solutions; it deals with feasibility and unboundedness.

   e. The theorem applies to all convex sets, including unbounded feasible regions.

2. Consider a primal linear programming problem and its dual. The primal problem is found to be infeasible. Using the Separating Hyperplane Theorem, what can we conclude about the dual problem?

   a. While both primal and dual can be infeasible, the Separating Hyperplane Theorem states that if one is infeasible, the other is unbounded (or infeasible).

   b. If the primal problem is infeasible, the Separating Hyperplane Theorem implies that the dual problem is unbounded (or infeasible). This is a fundamental result in linear programming duality.

   c. The dual problem might be unbounded, not necessarily have a unique optimal solution.

   d. The dual problem could be unbounded, but not necessarily feasible.

   e. The Separating Hyperplane Theorem provides a direct link between the primal and dual problem's feasibility and boundedness.

3. Suppose we have a primal linear programming problem and its dual. Both problems are feasible. What does strong duality tell us about the relationship between their optimal objective function values?

   a. This describes weak duality, which holds for any feasible solutions, not just optimal ones.

   b. This is also related to weak duality, but reversed for a maximization primal problem.

   c. Strong duality states that if both the primal and dual problems are feasible, then their optimal objective function values are equal. This is a fundamental result in linear programming.

   d. Strong duality establishes a precise relationship between the optimal values.

   e. This is incorrect; strong duality states equality, not strict inequality.

4. Explain the concept of complementary slackness in linear programming and its significance in verifying the optimality of solutions. Illustrate your explanation with a simple example.

   a. Complementary slackness is a fundamental concept in linear programming duality and is crucial for verifying optimality.

   b. This statement is incorrect; if a primal constraint is binding, the corresponding dual variable is *positive* (for a maximization problem), and vice versa.

c. Complementary slackness states that for optimal primal and dual solutions: If a primal variable is positive, the corresponding dual constraint is binding (holds with equality), and if a dual variable is positive, the corresponding primal constraint is binding. This provides a way to verify optimality. For example, consider a primal problem with a constraint $x_1 + x_2 \leq 5$ and its dual with a variable y_1. If $x_1 + x_2 = 5$ (binding), then $y_1 > 0$. If $y_1 = 0$, then $x_1 + x_2 < 5$ (non-binding).

d. Complementary slackness applies to problems with inequality constraints as well, after introducing slack variables.

e. Complementary slackness, along with primal and dual feasibility, is a *sufficient* condition for optimality.

5. Consider a primal linear programming problem and its dual. Explain the relationship between the feasibility and boundedness of the primal and dual problems, covering all four possible scenarios. For each scenario, describe the implications for the optimal objective function values of both problems.

a. This option incorrectly states relationships between primal and dual feasibility and boundedness. It doesn't account for strong duality when both are feasible and bounded.

b. This option is partially correct but doesn't fully capture the possibilities when the primal is infeasible. The dual could be infeasible or unbounded in that case.

c. This option incorrectly states that if the primal is infeasible and unbounded, the dual is feasible and bounded. This is not true.

d. Option D correctly describes the relationships. If the primal is feasible and bounded, strong duality guarantees the dual is feasible and bounded with equal optimal objective values. If the primal is infeasible, weak duality only tells us the dual is either infeasible or unbounded. If the primal is unbounded, the dual must be infeasible. Similarly, if the primal is infeasible, the dual is either infeasible or unbounded.

e. This option incorrectly describes the relationships between primal and dual feasibility and boundedness. It doesn't account for strong duality when both are feasible and bounded.

6. Given the primal problem: Maximize $Z = 2x_1 + 3x_2$ subject to $x_1 + x_2 \leq 5$, $x_1 \leq 3$, $x_2 \leq 4$, $x_1, x_2 \geq 0$. What is the objective function of its dual problem?

a. The dual problem is a minimization problem. The objective function coefficients are the right-hand side values of the primal constraints.

b. This is incorrect; the dual of a maximization problem is a minimization problem.

c. This is incorrect; the objective function coefficients should be the right-hand side values of the primal constraints.

d. This is incorrect; the objective function coefficients should be the right-hand side values of the primal constraints.

e. This is incorrect; the objective function coefficients should be the right-hand side values of the primal constraints.

7. For the primal problem: Maximize $Z = 3x_1 + 2x_2$ subject to $-x_1 + 3x_2 \leq 12$, $x_1 + x_2 \leq 8$, $2x_1 - x_2 \leq 10$, $x_1, x_2 \geq 0$, what are the constraints of its dual problem?

a. The coefficients of the dual constraints are the transpose of the primal constraint matrix, and the inequalities are reversed.

b. The inequalities should be $\geq$ for a minimization problem.

c. The constraints are missing the non-negativity conditions on the dual variables.

d. The coefficients of the dual constraints are incorrect. They should be the transpose of the primal constraint matrix.

e. The coefficients are incorrect, and the inequalities should be $\geq$ for a minimization problem.

8. In the context of weak duality, if Z represents the objective function value of the primal problem and W represents the objective function value of the dual problem, which inequality always holds for a maximization primal problem?

a. Weak duality states that the objective function value of any feasible solution to the primal problem is less than or equal to the objective function value of any feasible solution to the dual problem for a maximization primal problem.

b. This is incorrect. For a maximization primal problem, the primal objective function value is always less than or equal to the dual objective function value.

c. This is incorrect. This only holds true at optimality (strong duality).

d. This is incorrect. The inequality is $Z \leq W$ for a minimization primal problem and $Z \geq W$ for a maximization primal problem.

e. This is incorrect. The inequality is $Z \leq W$ for a minimization primal problem and $Z \geq W$ for a maximization primal problem.

9. Suppose the primal problem has an unbounded objective function. What can we conclude about the dual problem based on weak duality?

a. Incorrect. If the primal is unbounded, the dual is infeasible.

b. If the primal problem is unbounded, then the dual problem must be infeasible. This is a direct consequence of weak duality.

c. Incorrect. If the primal is unbounded, the dual is infeasible.

d. Incorrect. If the primal is unbounded, the dual is infeasible.

e. Incorrect. Weak duality provides a direct conclusion about the dual's feasibility.

10. Consider the primal problem: Maximize $Z = 3x_1 + 2x_2$ subject to $-x_1 + 3x_2 \leq 12$, $x_1 + x_2 \leq 8$, $2x_1 - x_2 \leq 10$, $x_1, x_2 \geq 0$. Formulate its dual problem. What is the constraint in the dual problem corresponding to the primal variable x_1?

a. The dual problem is formulated as follows: Minimize $W = 12y_1 + 8y_2 + 10y_3$ subject to $-y_1 + y_2 + 2y_3 \geq 3$, $3y_1 + y_2 - y_3 \geq 2$, $y_1, y_2, y_3 \geq 0$. The constraint corresponding to x_1 is $-y_1 + y_2 + 2y_3 \geq 3$.

b. This inequality is reversed. The coefficient of x_1 in the objective function is 3, and the constraint should be greater than or equal to 3.

c. This constraint incorrectly uses the coefficients from the primal constraints without considering the signs.

d. This constraint uses the coefficient of x_2 in the objective function instead of x_1.

e. This inequality is reversed and uses the coefficient of x_2 in the objective function instead of x_1.